

Oxygen-dependent intracellular killing: (1)

There is requirement of oxygen for such kind of intracellular killing of bacteria. It is also termed as "Respiratory Burst" as requirement of glucose and oxygen get increased after phagocytosis. As a result, oxygen containing bacteriocidal radicals are produced.

Oxygen-independent killing: (2)

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins. These hydrolytic proteins are bacteriocidal in nature according to their modes of action.

Immunodeficiency: (3)

The failure or deficiency of immune system is called as immunodeficiency. It is a state of complete absence of immunity or effective immune response.

Inflammation: (4)

Inflammation is the body's attempt to protect the body from harmful stimulus. Complex tissue response.

- * Part of body's immune system
- * For wound healing
- * Inflammatory responses: series of events for inflammation.
- * For clearance of infection.

Factors affect the antibody & antibody test: (5)

The good antigen and antibody reactions are known due to high specific and sensitive nature. Various factors involved which determine the nature of antigen and antibody reactions which are as follows:

- 1- Affinity
- 2- Avidity
- 3- Antigen & Antibody ratio
- 4- Physical form

Affinity Maturation: (6)

The affinity of antibodies depends upon the dosage of antigen. The low dosage of antigen causes increase in the affinity of antibodies which is called affinity maturation.

Three ways of immune response against Extracellular Pathogens: (7)

- 1- Neutralization 2- Opsonization 3- Complement activation.

Principle / Formation of Hybridoma (8)

Hybridoma technology fuses a B cell with a myeloma cell to create a hybridoma cell that has the ability to produce large quantities of identical monoclonal antibodies.

Usage of Monoclonal Antibodies: (9)

- * Monoclonal antibodies are used in medicine and research for prevention, diagnosis and treatment of disease.
- * Immunotyping of immune cells
- * Typing of various tumors (like) e.g. leukemia
- * Immunohistochemistry of solid tumors
- * Immunotherapy against tumor markers.
- * Use in fluorescence activated cell sorting (FACS).

T-cell Receptor: (10)

The T-cell receptor is a protein complex found on the surface of T-cells, or T-lymphocytes, that is responsible for recognizing fragments of antigen as peptides bound to major histocompatibility complex molecules.

V(D)J Recombination: (11)

Rearranges gene segments to generate diverse antigen receptors. It occurs in B and T cells during the development and is mediated by recombination-activating genes (RAG1 & RAG2).

B-Cell Receptors: (12)

composed of surface immunoglobulin molecule with a certain specificity for antigen. BCR can recognize following antigens in soluble form;

- * Nucleic acids
- * Polysaccharides
- * Some lipids
- * Small nucleotides (i.e. haptens)
- * Protein (both Native & denatured ^{determinants})

Structure of T-cell receptors: (13)

T-cell receptors are made up of two polypeptide chains that together composed of ~~(an~~^x alpha) one antigen binding region.

- * Approximately, 95% of TCRs are composed of an alpha and a beta chains, while the remaining 5% of TCRs are made up of Gamma and delta chains. There are multiple types of T-cells, and each has a specialized functions.

Difference b/w Antigen, Immunogen & Hapten: (14)

- * **Antigen:** A substance that reacts with the products of specific immune response.
- * **Immunogen:** A substance that induces specific immune system.
- * **Hapten:** A substance that is not immunogenic but can react with products of specific immune system.

Cellular Barriers: (15)

- * are physical barriers in the body that prevent the entry of pathogen.
- * made up of cells that are tightly packed together, preventing the passage of microbes.
- * Examples include epithelial cells lining the skin, respiratory tract and gastrointestinal tract as well as endothelial cells lining blood vessels.

Three accessory molecules involved in immune system response (16)
are CD4, CD8 & CD28. These molecules help regulate & enhance immune cell functions.

Categories of Cytokines (17)

Cytokines are categorized on following two basis.

1- Their source of origin; 2- Their functions

Monokines, Lymphokines, Interleukin^{1,2}
Chemokines

Mediators & regulators of Acquired immunity
3- Stimulators of Hematopoiesis

Innate immunity

Outcomes of Inflammation: (18)

In response of inflammatory process, following outcomes occur:

- * There is restoration of normal tissue
- * Large amount of tissue is destroyed which is called as fibrosis.
- * There is also a pus formation which is also termed as abscess formation.
- * Inflammation can be prolonged for longer time (periods) points which is also called chronic inflammation.

Define Isograft, Allograft & Xenograft: (19)

- * Isograft: This is a form of tissue which is transplanted between individuals of same genetic makeup like identical twin.
- * Allograft: This form of transplantation which can be from one person to other non-identical person.
- * Xenograft: This kind of transplantation is across the species for example from animal to human.

Diagnose of Type 1 Hypersensitivity: (20)

Type 1 hypersensitivity reactions are diagnosed in following imp ways;

- * By Skin (Prick & Intradermal) test observed in form of redness, itching and swelling of skin.
- * Measurement of total IgE levels (in patient's serum)
- * Allergen specific IgE levels

Drug Induced Hemolytic Anemia: (21)

* Also known as cytotoxic hypersensitivity * Effect various organs and tissues (Tissue specific) * Antigens are usually endogenous

* Exogenous: chemicals in form of haptens * bind to the surface of cell membrane * Antibodies against antigen: IgM or IgG

Examples of exogenous antigens: Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia.

* Ag-Ab reaction on the surface of RBC * Complement fixation on immune complex on RBC surface * Reaction time is minutes to hours * lysis of RBC.

Types of Autoimmunity: (22)

Classification based on tissues or organs involved two following categories;

* Organ specific Autoimmunity * Non-organ specific autoimmunity

Natural and Artificial active Immunizations: (23)

Natural active immunization occurs when the body develops immunity after being exposed to a pathogen naturally, such as through infection. Artificial active immunity is when immunity is induced through vaccination with weakened or inactivated pathogens.

How antibodies responds: (24)

Antibodies responds by binding to specific antigens, making them for destruction by immune cells.

Functions of Artificial active Immunology: (25)

Artificial active immunization function by stimulating the immune system with vaccines to produce a protective response against specific pathogen.

Effector function of Complement Proteins: (26)

- * Opsonization (Opsonins) * Chemoattraction (chemoattractants)
- * Anaphylaxis (Anaphylaxins) * Pro-inflammatory

How antibodies are formed: (27)

Antibodies are formed when B-cells in the immune system encounter antigens, triggering their differentiation into plasma cells that produce specific antibodies.

Hypersensitivity: (28)

refers to an exaggerated immune response to harmless substances, resulting in allergies or autoimmune disease.

MHC Class II loci Names: (29)

- * HLA-DP * HLA-DQ * HLA-DR

Autoimmunity: (30)

refers to a condition where the immune system mistakenly attacks and damages the body's own tissues and organs. It can lead to various autoimmune disease.

B-cell structure: (31)

B-cells are a type of white blood cell involved in immune (system) response. They have a membrane-bound antibody on their surface called B-cell receptor (BCR) that recognize specific antigens.

Phagocytosis: (32)

is a process where specialized cells called phagocytes (They have a membrane) engulf and ~~respond~~ destroy foreign particles, such as bacteria and debris.

Transplant response against immune System: (33)

The immune system can mount a response against transplanted organ, leading to rejection. Immunosuppressive medications are used to prevent this response.

PAMPs (Pathogen-Associated Molecular Patterns) (34)

PAMPs are molecules found on pathogens that trigger an immune response. An example of a PAMP is lipopolysaccharide (LPS) found on the outer membrane of certain bacteria.

Chronic Rejection: (35)

refers to a long-term immune response against a transplanted organ, leading to its gradual deterioration.

Organ Specific and Non-organ specific autoimmunity. (36)

*Organ-specific autoimmunity refers to autoimmune diseases where the immune system attacks specific organs, such as the thyroid in Hashimoto's thyroiditis. Target organ; Skin, Thyroid glands, Muscles

*Non-organ specific autoimmunity refers to autoimmune diseases that affect multiple organs or systems, like systematic lupus erythematosus.

Target organs; skin, joints, soft tissues

Anatomical & Physical barriers of Immune System. (37)

like skin, mucous membranes, and cilia that helps prevent pathogens from entering the body.

Opsonization: (38)

is when molecules mark pathogens for immune cell elimination, aiding in their engulfment and destruction.

Immunity against tumor cells: (39)

The immune system can develop immunity against tumor cells through mechanisms like cytotoxic T cells & natural killer cells.

Primary function of Immunoglobulin antibodies: (40)

Immunoglobulins primarily function by binding to specific antigens using paratopes, determining their specificity. The valency of an antibody refers to the number of available antigen binding sites, with a minimum valency of two.

Complement proteins removes extracellular pathogens from blood: (41)

Complement proteins mark pathogens for engulfment, create pores in pathogen membranes to destroy them, and activate the adaptive immune response if needed.

When virus or bacteria enter a healthy body, the immune system activates physical barriers, immune cells, antibodies and inflammation to neutralize & eliminate the invaders, safeguarding the body from infection.

Domains of Immunoglobulin Structures: (42)

Domains are folded regions which contain an intra-chain disulfide bond. Usually, heavy chains contain four domains while light chains composed of two domains.

Role of Chemical and Biological Adjuvants: (43)

Chemical Adjuvants: * Aluminium salt (Alum)

* Only chemical suitable for human use * Used in DTP (Diphtheria, Tetanus & Pertussis)

Role of Adjuvants in immunization.
* Alum causes slow release of antigen * Increase TLR interaction
* Increase the activation of mononuclear phagocytes * Also induce the increased cytokine secretion.

Biological Adjuvants: * Certain bacteria used as biological adjuvant

* Bacterial products are used in combination with oil & detergent.
* Activation of mononuclear phagocytes & cytokine secretion.

Complement Activation: (45)

Complement proteins are activated through a series of cascading reactions which are called as complement activation. Imp pathways of C.A;

- 1) Classical Pathway 2) Alternate Pathway 3) Mannose Binding lectin (MBL) Pathway

Cytokines: (46)

Cytokines are secretory molecules which act as mediators b/w immune cells. IL-1 induces fever and the production of acute phase proteins, some of which are antimicrobial because they can opsonize bacteria.

Primary lymphoid organs (47)

- * Generative & central lymphoid organs
- * Involved in maturation of lymphocytes
- * Contain stem cell for division and maturation
- * Bone marrow, Thymus

Secondary lymphoid organs

- * Peripheral lymphoid organ
- * Involved in providing adaptive immune system
- * Site of interaction of antigen, APC and lymphocytes
- * Various anatomical location

lymph nodes
spleen
Tonsils

MHC I and MHC II: (48)

are proteins that play a crucial role in the immune system.

MHC I

- * MHC I molecules present antigens from inside the cell to cytotoxic T cells
- * Structure: MHC I molecules consist of one membrane-spanning α chain produced by MHC genes, and one β chain produced by β 2-microglobulin gene

MHC II

- * MHC II molecules present antigen from outside the cell to ^{helper} T cells.
- * Structure: MHC II molecules consist of two membrane-spanning α and β both produced by MHC genes

Onco-fetal Antigens: (49)

Onco-fetal antigens ^{Present} proteins during fetal development and reappear in some tumor cells (cancers). They help detect and monitor cancer and can be targeted for treatments.

DiGeorge Syndrome: (50)

is a genetic disorder caused by a deletion on chromosome 22, leading to heart defects, immune system issues, facial and developmental challenges, and more. Early diagnosis and specialized care are vital.

Antibody response against T-cell antigen: (51)

Antibodies are produced by B cells and target extracellular pathogens. T cells recognize intracellular antigens and coordinate immune response. T cells can also assist B cell in generating antibodies when needed, ensuring a comprehensive immune defense.

Role of immune system: (52)

The immune system defends against threats, remember them, and maintains health through recognition, response, regulation and adaptation.

T cell Activation: (53)

T cell activation involves antigen recognition, co-stimulation, intracellular signaling, clonal expansion, effector functions and memory formation.